Syllabus

Module I (11 Hours)

Representation of signed numbers – 1's complement and 2's complement ,Logic gates – AND - OR – NOT - NAND- NOR - XOR , Boolean algebra - Basic laws and theorems , Boolean functions - truth table, Standard forms of Boolean Expressions – Sum of Products and Product of Sums - minimization of Boolean function using Karnaugh map method - Realization using logic gates, Floating point numbers

Combinational Circuits - Half adder - Full Adder- Decoder -Encoder- Multiplexer – Demultiplexer

Module II (10 Hours)

Sequential circuit - Clocking, Flip flops - SR – JK- D -T flip flops, Counters - Synchronous and asynchronous counters - UP/DOWN counters , Registers - Serial in serial out - Serial in parallel out - Parallel in serial out - Parallel in parallel out registers

A practical assignments may be given in configuring a PC / configuring arduino - Implementing simple programs for blinking an LED - Input from an external switch - fading an LED - serial monitor and debugging / installing & configuring Raspberry pi.

Module III (10 Hours)

Computer abstractions and technology - Introduction, Computer architecture -8 Design features, Application program - layers of abstraction, Five key components of a computer, Technologies for building processors and memory, Performance, Instruction set principles – Introduction, Classifying instruction set architectures, Memory addressing, Encoding an instruction set.

Module IV (9 Hours)

The Processor - Introduction, Logic design conventions, Building a datapath, A simple implementation scheme, An overview of pipelining - Pipelined datapath and control - Structural hazards - Data hazards - Control hazards

I/O organization - Accessing I/O devices, interrupts - handling multiple devices, Direct memory access

Programming assignments may be given in any HDL like Verilog or VHDL to create gate level/ Dataflow/Behavioural level models of gates, multiplexers, adders, flip-flops, registers etc. No detailed teaching of HDL is necessary. The students can be given a basic tutorial write up on gate level modelling.

Module V (8 Hours)

The Memory System – basic concepts, semiconductor RAM memories - organization – static and dynamic RAM, Structure of larger memories, semiconductor ROM memories, Speed, Size and cost ,Cache memory – mapping functions – replacement algorithms , Virtual memory – paging and segmentation.